ANA 222, Introduction to thorax and typical intercostal space

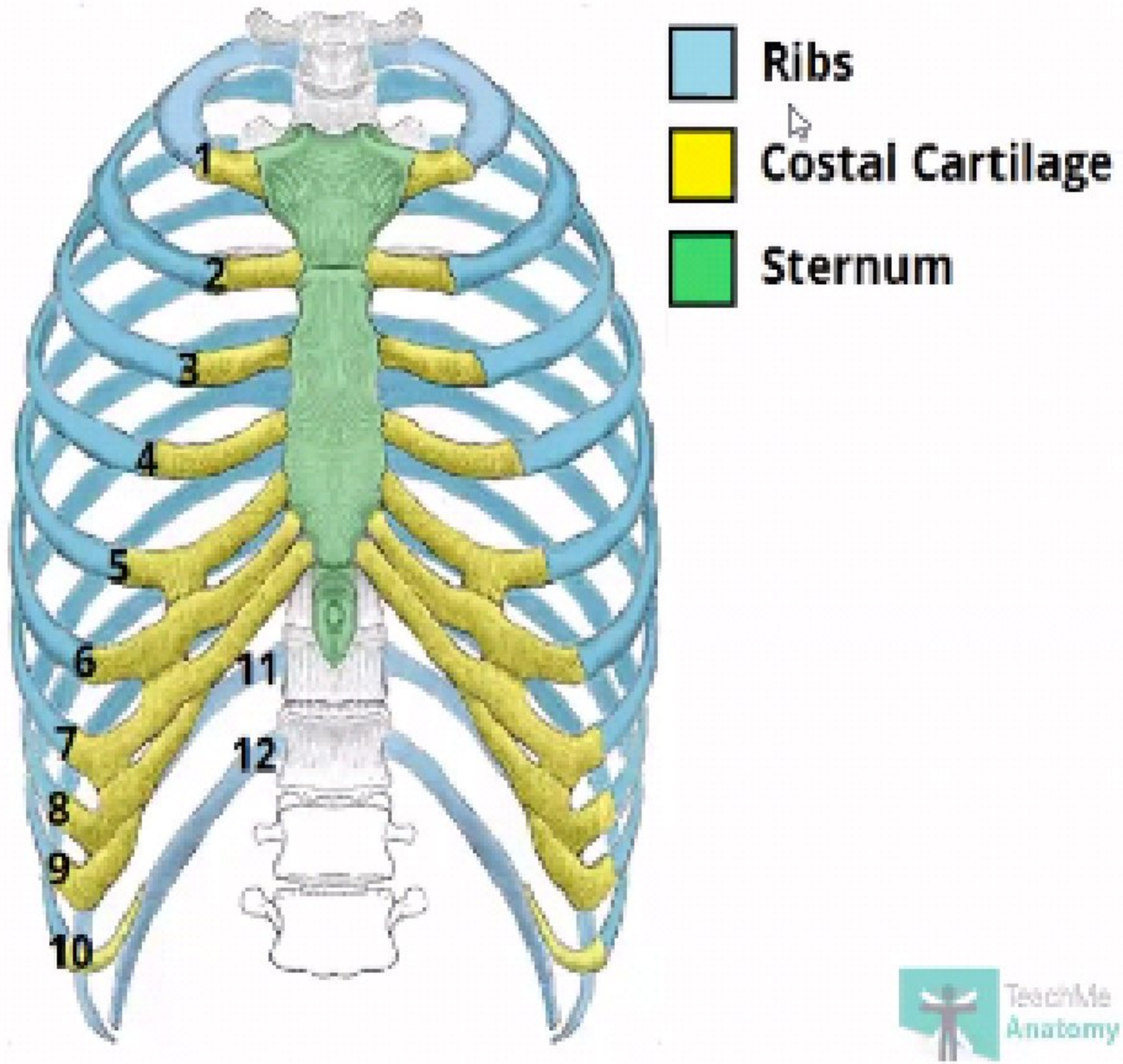
By Dr. Mbaka G.O.

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Thorax is a cavity of osseous and cartilages which provide protection and support for heart, great vessels and lungs.

Boundaries:

- Anteriorly by sternum and cartilages.
- Laterally by ribs.
- Posteriorly by 12 thoracic vertebral.
- Thorax is roofed by supra-pleural membrane and floored by respiratory diaphragm.



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Ribs: The ribs that directly articulates with sternum are true ribs.

The ribs that do not have direct articulation with the sternum are false ribs.

Ribs 1-7 are true ribs, 8-10 false ribs and 11-12 floating ribs.

There could be the absence of 1 or additional one rib.

Rib bones arching below the other rib is separated by gaps-intercostal spaces. The spaces are deeper in front than behind and decreased superior inferiorly.

- Ribs usually are oblique. The upper two are less oblique. The obliquity increases and becomes maximum at the 9th rib.
- The rib length increases from 1-7th and then decreases afterward.
- The breadth decreases from the 1st rib down.

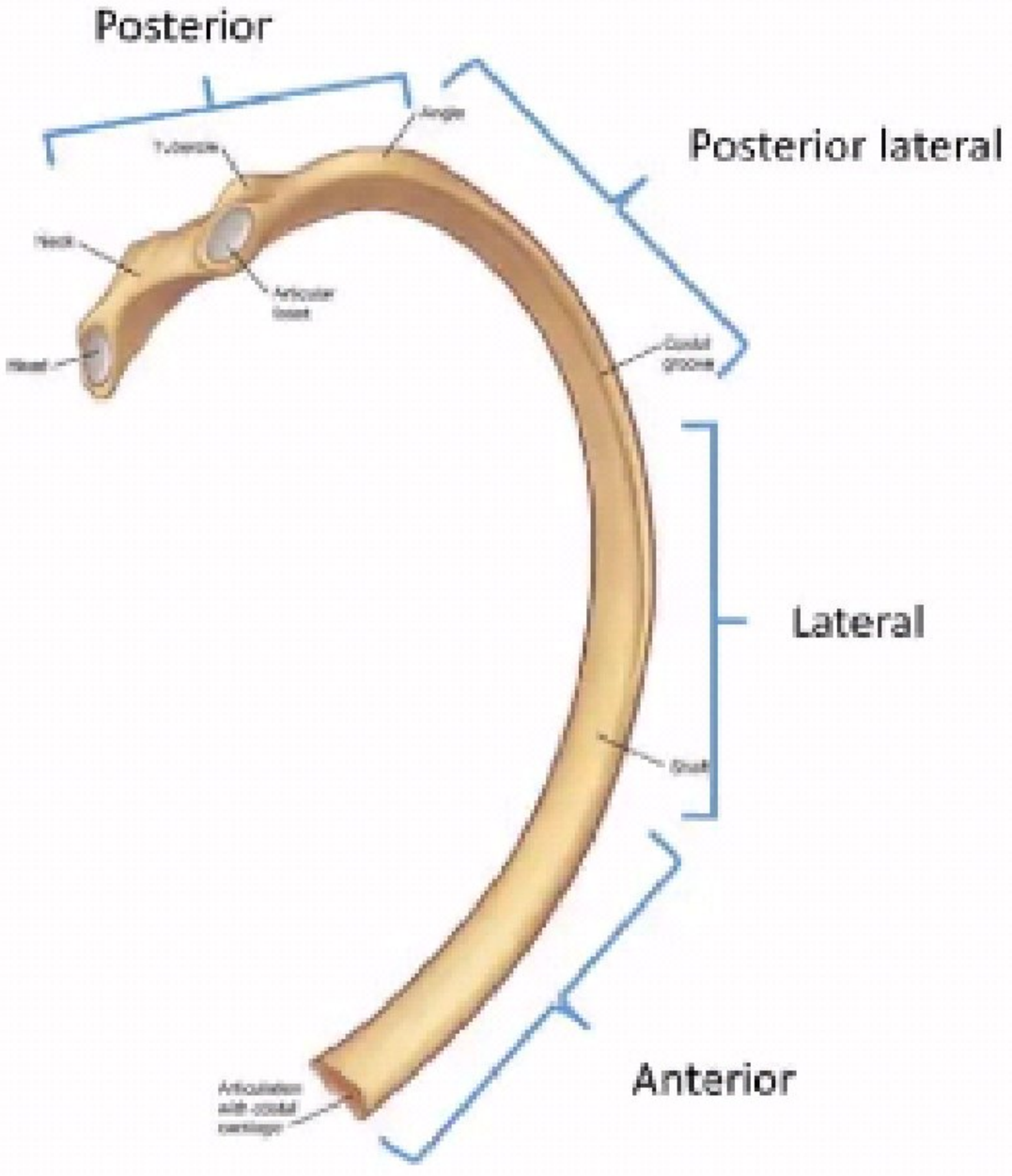
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Typical ribs: 3-9 ribs are typical ribs while the remaining are not typical ribs.

Characteristics of typical ribs,

Head: On the head are two facets separated by a crest. The lower facet articulates with own vert.

Neck: It is in front of transverse process of own vert. There are 2 tubercles in the junction between the neck and shaft. The lateral part is non articular while the medial part form costotransverse junction of own vert.



Shaft: It is flat with 2 surfaces and groove under. It is bent at an angle 5cm lateral to the tubercle.

Ossifications: 4 centres- body, head, articular and non articular part of tubercles.

From shaft 8wks, head and tubercles 16th 20th years and unifying with the body 25yrs of age.

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Typical intercostal spaces:

It contains three incomplete layers of muscle.

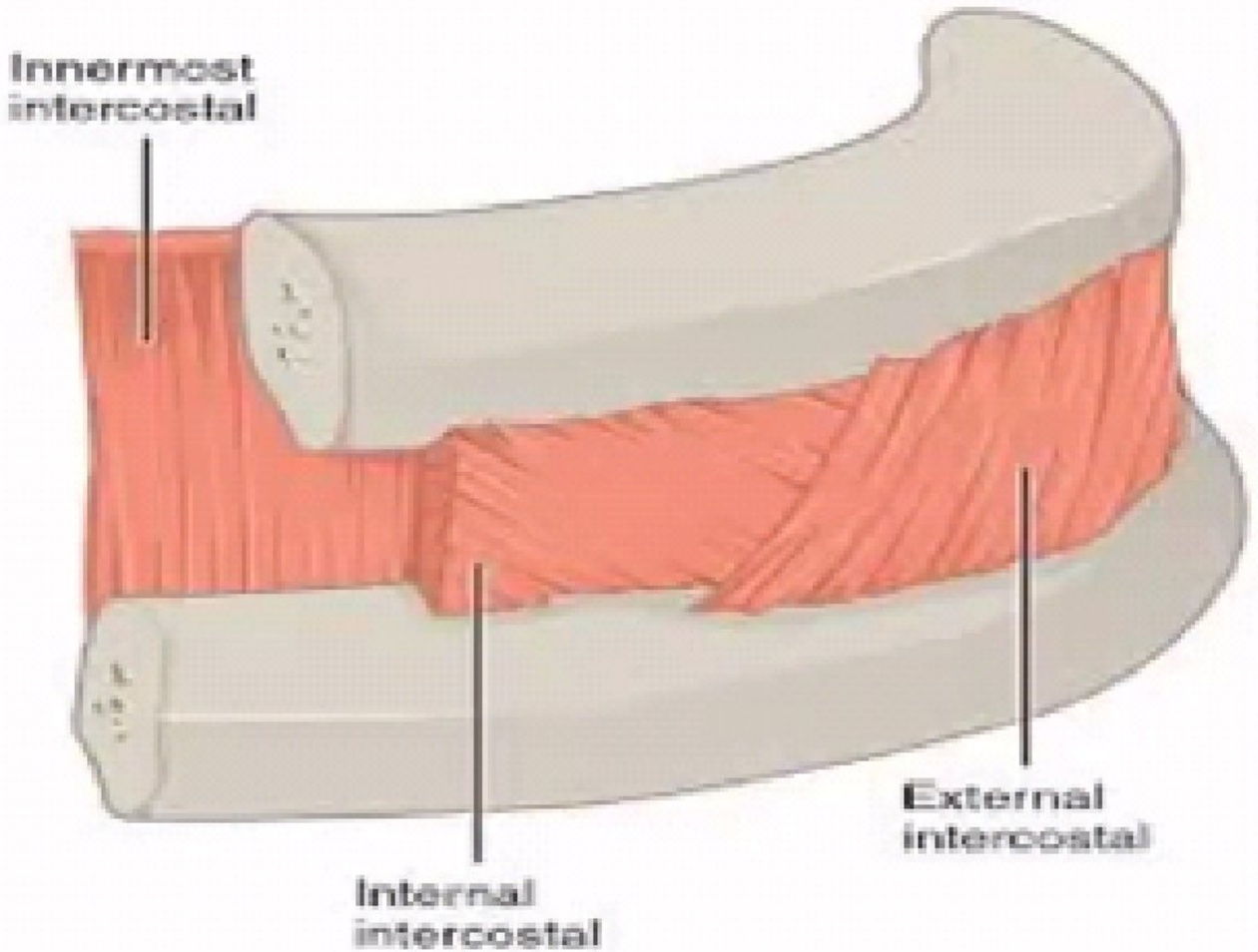
External, internal and innermost intercostal muscles, all concern in costal movements and involved in respiration.

External intercostal: From the transverse of rib posterior close to superior costotransverse lig. Extends almost to costochondrial junction and replaced by external intercostal membrane from costal cartilage to the sternum. The fibres are thicker than that of internal. It runs anterior inferiorly. They are 11 pairs extending from the lower border of upper rib to upper of lower ribs.

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Internal intercostal: They are 11 pairs, Extends from the margins of sternum to angle of ribs. Posterior to it is replaced by internal intercostal membrane. Each arises from costal groove above and adjacent costal cart above to upper border of rib below. Fibres run posterior inferior which is at right angle to the external intercostal.

Innermost intercostal: It has variable extent usually it crosses more than one space and extends further posterior than internal intercostal but differentiated from it by the presence of intercostal nerve & vessels. The vein most prox than artery & nerve (VAN).



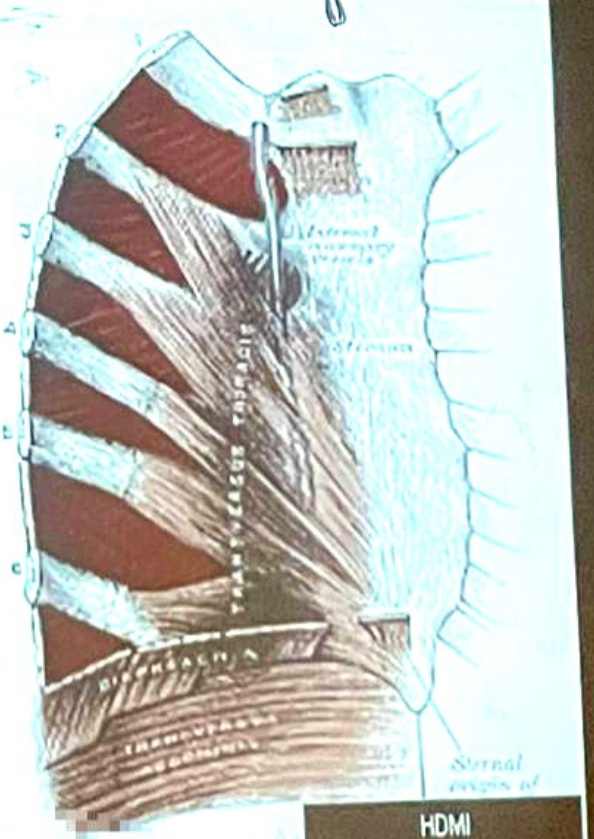
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Transversus thoracic: It arises from the lower end of sternum from where the digitations diverge on each side of costal cartilage from the sixth up to the second cartilage.

Nerve supply: Intercostal nerves.

Action: External intercostal active during inspiration and shortens.

Internal intercostal and innermost intercostal shortens during expiration.



The neurovascular bundle is more proximal within the intercostal spaces.

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Costochondritis and Tietze syndrome: Inflammation of the cartilage connecting the rib to the sternum will result to chest pain. In Tietze syndrome, the affected cartilage is swollen usually at the connecting point with the sternum.

Clinical Relevance: Rib Fractures

Rib fractures most commonly occur in the middle ribs, as a consequence of **crushing** injuries or direct trauma. A common complication of a rib fracture is further **soft tissue injury** from the broken fragments. Structures most at risk of damage are the lungs, spleen or diaphragm.

If two or more fractures occur in two or more adjacent ribs, the affected area is no longer under control of the thoracic muscles. It displays a paradoxical movement during lung inflation and deflation. This condition is known as **flail chest**. It impairs full expansion of the ribcage, thus affecting the oxygen content of the blood. Flail chest is treated by fixing the affected ribs, preventing their paradoxical movement.

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Congenital deformities: Pectus Excavatum; sunken or depressed sternum.

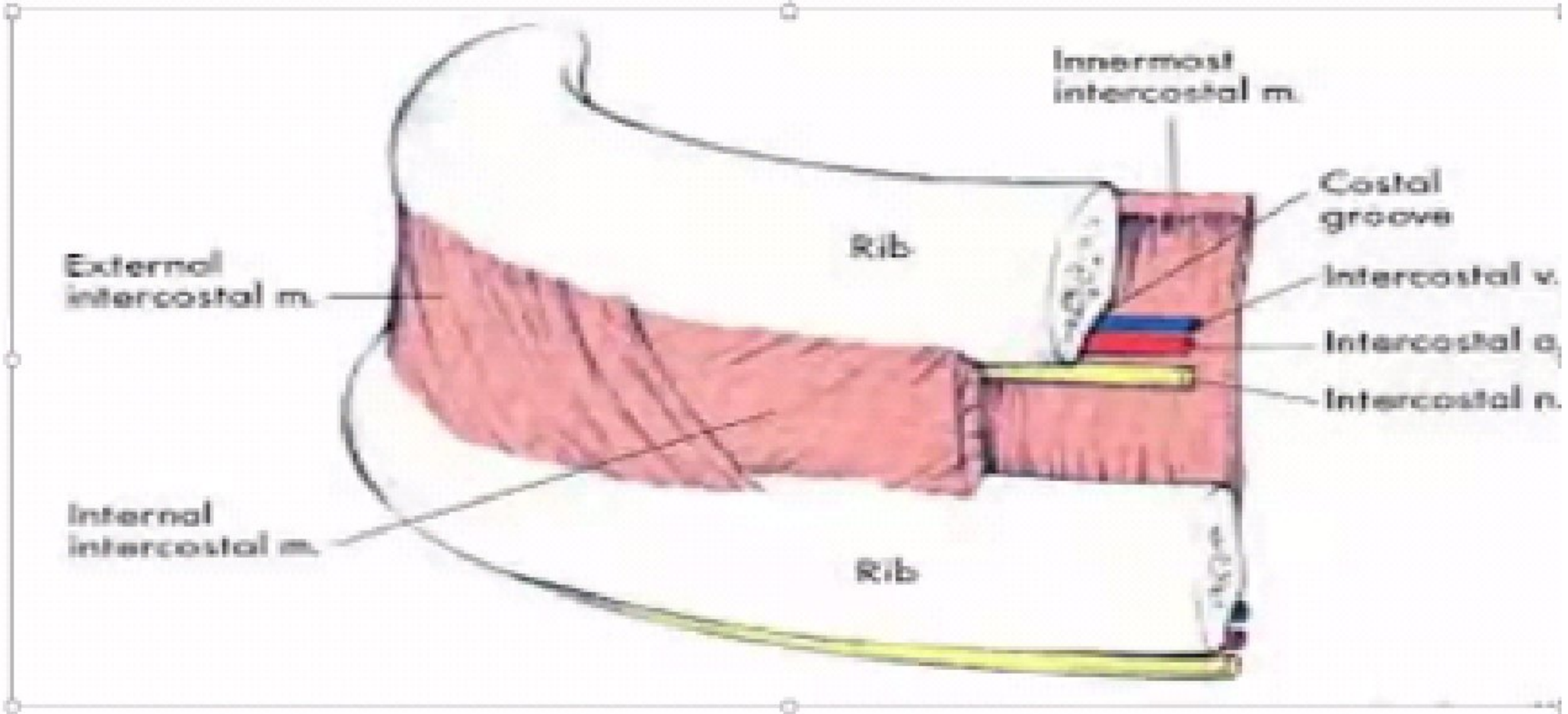
Pectus carinatum; protruding sternum.

Intercostal hernia: A defect in the intercostal muscle, secondary to trauma, surgery, and congenital defects, may result in prolapse (or bulging) of the lung or abdominal organs. The latter usually requires damage to the diaphragm as well. The herniation may cause pain, or become incarcerated, or strangled, resulting in tissue necrosis.

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